# Alif Tasbir

aliftasbir@gmail.com | Leicester, UK | github.com/a1if | a1if.github.io

## PROFILE

MSc Artificial Intelligence candidate at De Montfort University who designs, fine-tunes and deploys large language model (LLM) and retrieval-augmented generation (RAG) applications in Python. Hands-on with PyTorch, QLoRA fine-tuning, FAISS and ChromaDB vector search and FastAPI services, with published deep learning research in clinical AI. Seeking an AI/ML Engineer role to build and deploy machine learning solutions for real business problems.

## EDUCATION

**MSc Artificial Intelligence**, De Montfort University, Leicester, UK | Sep 2025 to Sep 2026 (expected)

Modules: Neural Systems and Natural Language Processing; Fuzzy Logic and Evolutionary Computing.

**BSc Computer Science and Engineering**, University of Information Technology and Sciences (UITS), Dhaka, Bangladesh | Aug 2018 to Aug 2022

CGPA 3.74/4.00. Modules: Machine Learning, Data Structures and Algorithms, Software Engineering, Object Oriented Programming.

## TECHNICAL SKILLS

- **Languages:** Python, C, C++, Java, JavaScript, HTML, CSS
- **Machine learning and deep learning:** PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, CNNs, LSTMs
- **LLMs and RAG:** LLM fine-tuning (QLoRA, PEFT, TRL), Hugging Face Transformers, retrieval-augmented generation (RAG), LangChain, LangGraph, Ollama, Sentence Transformers, vector databases (FAISS, ChromaDB), LLM evaluation (BLEU, ROUGE, BERTScore), Gemini API, natural language processing (NLP)
- **APIs and apps:** FastAPI, Flask, Django, Streamlit, Gradio
- **MLOps and tools:** Git, MLflow, DVC, Docker
- **Data:** SQL (MySQL), NoSQL, Excel

## SELECTED PROJECTS

**QLoRA Fine-Tuning and Multi-Hop RAG Pipeline** | Python, PyTorch, PEFT, TRL, FAISS, Sentence Transformers, Gradio

- Fine-tuned DeepSeek-R1-Distill-Qwen-1.5B with QLoRA (4-bit NF4, r=128) on 2,000 Alpaca samples, covering the full SFT pipeline and adapter merging.
- Built a multi-hop RAG pipeline over 5,000 arXiv abstracts using FAISS, Sentence Transformers and sub-question decomposition for complex queries.
- Evaluated base, fine-tuned and RAG models with BLEU, ROUGE and BERTScore, and deployed a Gradio chat interface with source attribution.

**Multimodal RAG Application** | Python, FastAPI, Streamlit, FAISS, Gemini 2.5 Flash, gpt-oss-20b

- Built a bilingual (Bangla and English) RAG system on a custom FAISS vector store for semantic retrieval.
- Designed a FastAPI backend supporting real-time web search and an adaptive deep-reasoning mode.
- Added dynamic routing between Gemini 2.5 Flash and gpt-oss-20b to balance answer quality against API rate limits.

**Heterogeneous Multi-Agent Edge AI for Clinical Decision Support** (MSc dissertation)

- Built a two-agent ECG pipeline: a CNN-LSTM heartbeat classifier sends events to a frozen Gemma 4 E4B model, either as text or through a trained linear adapter that emits 4 virtual tokens per event.
- Cut the language model's prompt-processing time by 24–78% against compact text, and improved balanced accuracy at 10 and 20 events (0.83 vs 0.73), with held-out patients, three training seeds and patient-level bootstrap statistics.
- Ran batched serving on a 12 GB GPU: 10–30% higher median throughput and 12–28% lower median energy per decision than filtered text. Measured energy with NVML power sampling at 100 Hz.

## WORK EXPERIENCE

**Junior Filing Specialist**, Compliance Services Limited (part of WSD Group), Dhaka, Bangladesh | Apr 2024 to Sep 2025

- Handled live filings and manually proofread documents to verify accuracy.
- Converted Word documents to HTML and built webpages using HTML and CSS.
- Maintained filing records in Excel, kept documentation up to date and analysed data.

## RESEARCH EXPERIENCE

**Research Assistant**, University of Information Technology and Sciences (UITS), Dhaka, Bangladesh | Jul 2022 to Mar 2023

- Developed graph-based deep learning models for epileptic seizure detection from EEG signals, improving detection accuracy.
- Applied signal processing, CNNs and statistical inference to optimise model results.
- Contributed to grant proposals and experiment design.

## PUBLICATIONS

1. Epileptic seizure detection from electroencephalogram (EEG) signals using linear graph convolutional network and DenseNet based hybrid framework. ScienceDirect, Elsevier.
2. Parkinson's disease detection from EEG signal employing autoencoder and RBFNN-based hybrid deep learning framework utilizing power spectral density. Digital Health, Sage Journals.